Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence and Data Science

Name: Krish Makhija	Roll no: 57	Branch: AI&DS
Subject: ASD&D	Batch: T13	Date: 30-07-2026

EXPERIMENT 2: To Containerize Application using Docker

This experiment maps to LO4.

LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory:

Installation and setup:

To execute containerized workloads locally, the Docker engine and tooling must be downloaded and installed on the host operating system:

- **Docker Desktop (Windows & macOS):** An all-in-one GUI and CLI application that packages the Docker Daemon, Docker CLI, Docker Compose, Buildx, and a local Kubernetes environment.
    - **Prerequisites for Windows:** Requires enabling hardware virtualization in BIOS and setting up **WSL2 (Windows Subsystem for Linux 2)** as the underlying Linux kernel backend.
    - **Installation Workflow:** Download the installer from the official Docker website (`docker.com`), run the setup wizard, enable the WSL2 backend component, and reboot the system.
- **Docker Engine (Linux):** On Linux distributions (e.g., Ubuntu, Debian, CentOS), Docker is installed directly via the system package manager (`apt` or `yum`) by adding Docker's official repository, installing `docker-ce` (Community Edition), and enabling the `dockerd` system service via `systemctl`.
- **Post-Installation Verification:** Once installed, launching the terminal and running basic verification commands (such as `docker --version` or running the lightweight `hello-world` test container) verifies that the client can successfully establish communication with the background daemon process.

Introduction:

- **Docker:** An open-source platform designed to automate application deployment using OS-level virtualization through containers.
- **Containerization:** A lightweight alternative to Virtual Machines (VMs) that packages code and dependencies together. Unlike VMs, containers share the host operating system kernel, making them significantly faster and less resource-intensive.
- **Docker Image:** A read-only, immutable template containing application code, system tools, libraries, and runtime configurations. It acts as the static blueprint for an application stack.
- **Docker Container:** A runnable, stateful instance created from a Docker Image. When launched, Docker adds a thin read-write filesystem layer on top of the immutable image, allowing active execution and resource allocation.

Architecture:

- **Docker Daemon (`dockerd`):** The persistent background service running on the host OS that builds, executes, and manages Docker objects such as images, containers, networks, and storage volumes.
- **Docker Client (`docker`):** The primary command-line interface (CLI) tool that allows users to interact with the daemon by sending REST API requests.
- **Docker Hub / Registry:** A centralized hosted service provided by Docker for finding, storing, and sharing public or private container images.

Practical Execution and Command Line Analysis:

System Environment Verification:

- `docker --version`: Queries the installed client version to confirm engine compatibility. In this experiment, the environment verified Docker version **25.0.3, build 4debf41**.
- `docker info`: Fetches a comprehensive system-wide diagnostic report. The execution logged host statistics including **31 total containers** (2 active, 29 stopped), **26 local images**, WSL2 kernel integration, allocated hardware resources (**20 CPUs, 15.5 GB RAM**), and active CLI plugins like Compose, Buildx, and Scout.

Image Repository and Search Management:

- **`docker images`**: Scans local engine storage and lists cached base images along with metadata such as repository tags, unique Image IDs, creation timestamps, and virtual disk sizes (e.g., `asd_exp3-backend`, `mysql`, `ubuntu`).
- **`docker search ubuntu`**: Queries the remote Docker Hub registry for public repositories matching the search string, returning descriptions, star ratings, and official build tags to evaluate image authenticity.
- **`docker pull ubuntu`**: Downloads the target base image layers from Docker Hub into local host storage, automatically applying the default: latest tag when no specific version is specified.

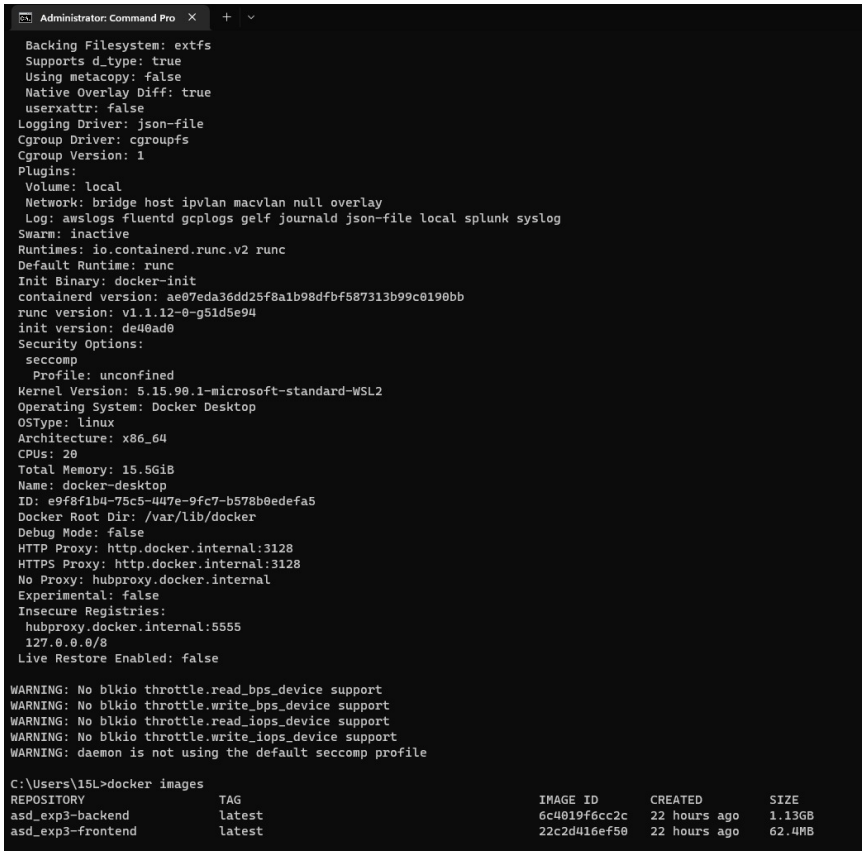
Container Execution and interactive Shell:

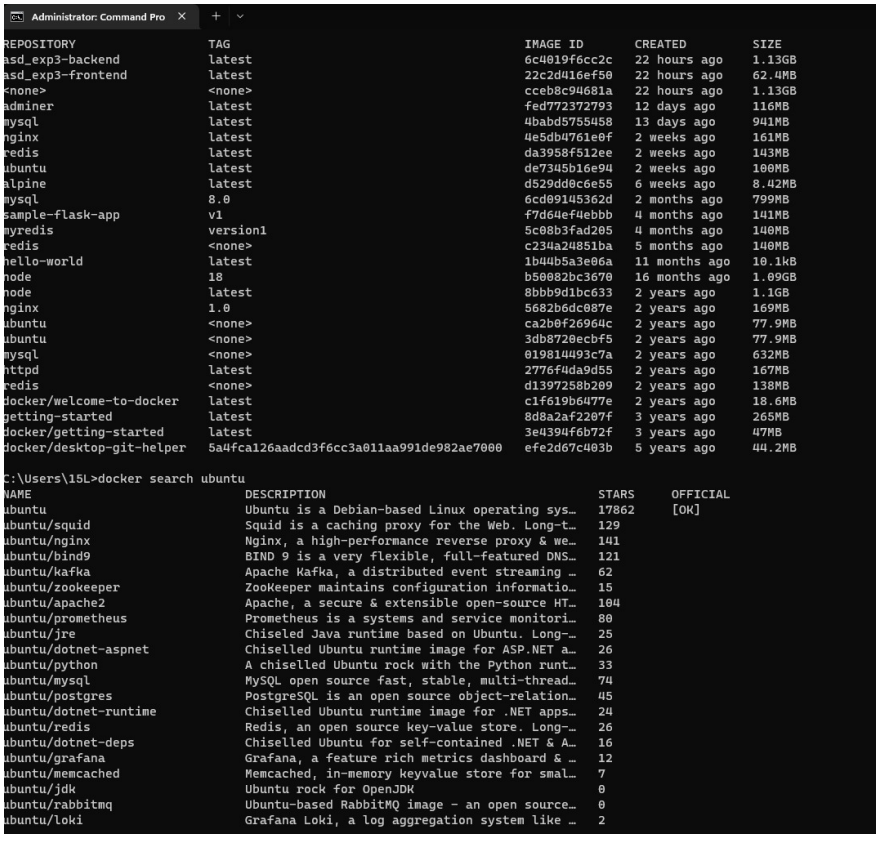
- **`docker run -it ubuntu`**: Instantiates and boots a new isolated container directly from the base `ubuntu` image.
    - **`-i` (Interactive):** Retains standard input (`STDIN`) open to continuously send terminal input into the container.
    - **`-t` (TTY):** Allocates a pseudo-terminal session, attaching the user's terminal to the container's bash shell (`root@<container_id>:/#`).
- **`pwd`** / **`ls`**: Commands executed directly inside the container instance to inspect the working directory and confirm the presence of an isolated root filesystem structure separate from the host.
- **`exit`**: Sends a termination signal to the interactive shell process, gracefully stopping the container execution.

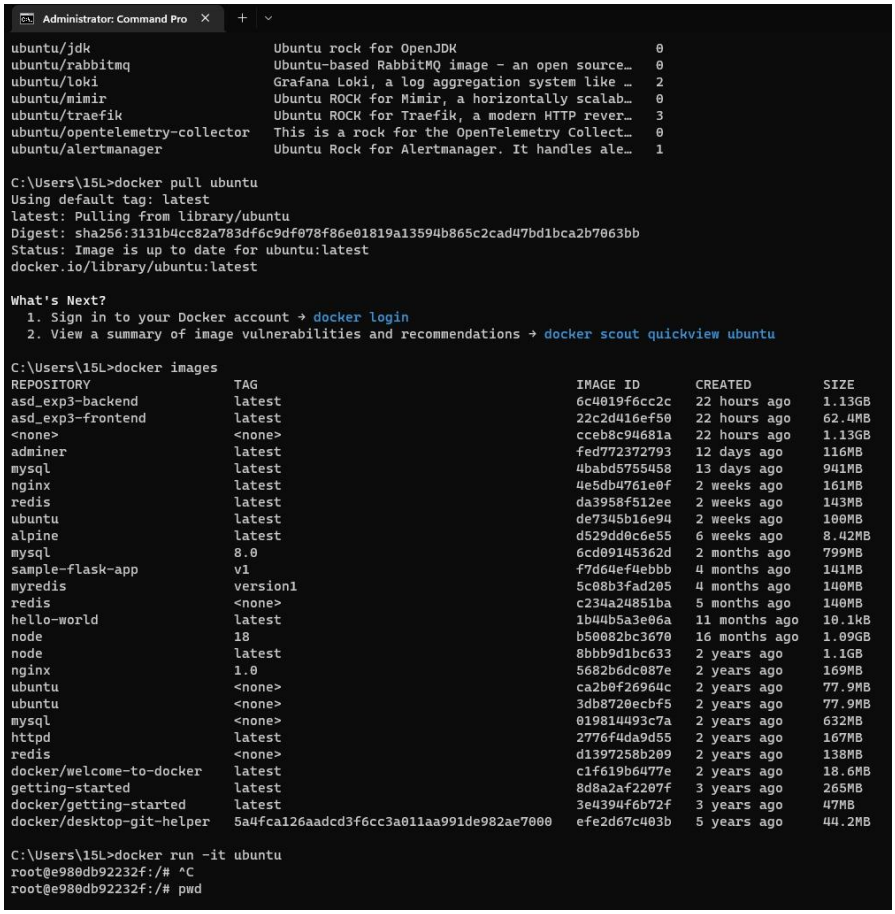
Lifecycle and Process Monitoring:

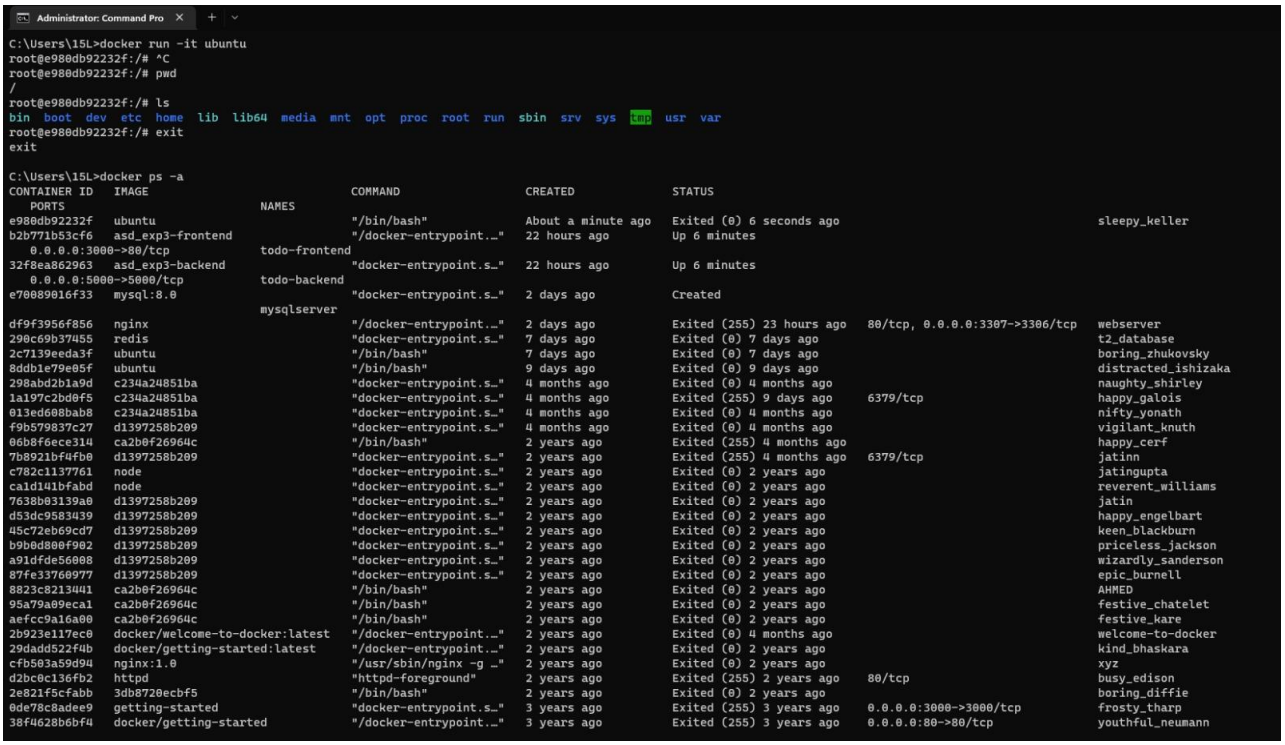
- **`docker ps -a`**: Queries the Docker daemon to list all container instances across all states (Running, Exited, Created). The terminal output provided detailed operational parameters, including active port mappings (such as `3000->80/tcp` for `todo-frontend`), container runtime durations, status codes (`Exited (0)`), and auto-generated container names (e.g., `sleepy_keller`).


Administrator: Command Pro
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.
C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41
C:\Users\15L>docker info
Client:
Version: 25.0.3
Context: default
Debug Mode: false
Plugins:
buildx: Docker Buildx (Docker Inc.)
Version: v0.12.1-desktop.4
Path: C:\Program Files\Docker\cli-plugins\docker-buildx.exe
compose: Docker Compose (Docker Inc.)
Version: v2.24.6-desktop.1
Path: C:\Program Files\Docker\cli-plugins\docker-compose.exe
debug: Get a shell into any image or container. (Docker Inc.)
Version: 0.0.24
Path: C:\Program Files\Docker\cli-plugins\docker-debug.exe
dev: Docker Dev Environments (Docker Inc.)
Version: v0.1.0
Path: C:\Program Files\Docker\cli-plugins\docker-dev.exe
extension: Manages Docker extensions (Docker Inc.)
Version: v0.2.22
Path: C:\Program Files\Docker\cli-plugins\docker-extension.exe
feedback: Provide feedback, right in your terminal! (Docker Inc.)
Version: v1.0.4
Path: C:\Program Files\Docker\cli-plugins\docker-feedback.exe
init: Creates Docker-related starter files for your project (Docker Inc.)
Version: v1.0.1
Path: C:\Program Files\Docker\cli-plugins\docker-init.exe
sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
Version: 0.6.0
Path: C:\Program Files\Docker\cli-plugins\docker-sbom.exe
scout: Docker Scout (Docker Inc.)
Version: v1.5.0
Path: C:\Program Files\Docker\cli-plugins\docker-scout.exe
Server:
Containers: 31
Running: 2
Paused: 0
Stopped: 29
Images: 26
Server Version: 25.0.3
Storage Driver: overlay2
Backing Filesystem: extfs
Supports d_type: true
Using metacopy: false


Administrator: Command Pro
Backing Filesystem: extfs
Supports d_type: true
Using metacopy: false
Native Overlay Diff: true
userxattr: false
Logging Driver: json-file
Cgroup Driver: cgroupfs
Cgroup Version: 1
Plugins:
Volume: local
Network: bridge host ipvlan macvlan null overlay
Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
Swarm: inactive
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version: ae07eda36dd25f8a1b98dfbf587313b99c0190bb
runc version: v1.1.12-0-g51d5e94
init version: de40ad0
Security Options:
seccomp
Profile: unconfined
Kernel Version: 5.15.90.1-microsoft-standard-WSL2
Operating System: Docker Desktop
OSType: linux
Architecture: x86_64
CPUs: 20
Total Memory: 15.5GiB
Name: docker-desktop
ID: e9f8f1b4-75c5-447e-9fc7-b578b0edefa5
Docker Root Dir: /var/lib/docker
Debug Mode: false
HTTP Proxy: http.docker.internal:3128
HTTPS Proxy: http.docker.internal:3128
No Proxy: hubproxy.docker.internal
Experimental: false
Insecure Registries:
hubproxy.docker.internal:5555
127.0.0.0/8
Live Restore Enabled: false
WARNING: No blkio throttle.read_bps_device support
WARNING: No blkio throttle.write_bps_device support
WARNING: No blkio throttle.read_iops_device support
WARNING: No blkio throttle.write_iops_device support
WARNING: daemon is not using the default seccomp profile
C:\Users\15L>docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
asd_exp3-backend latest 6c4019f6cc2c 22 hours ago 1.13GB
asd_exp3-frontend latest 22c2d416ef50 22 hours ago 62.4MB


Administrator: Command Pro
REPOSITORY TAG IMAGE ID CREATED SIZE
asd_exp3-backend latest 6c4019f6cc2c 22 hours ago 1.13GB
asd_exp3-frontend latest 22c2d416ef50 22 hours ago 62.4MB
<none> <none> cceb8c94681a 22 hours ago 1.13GB
adminer latest fed772372793 12 days ago 116MB
mysql latest 4babd5755458 13 days ago 941MB
nginx latest 4e5db4761e0f 2 weeks ago 161MB
redis latest da3958f512ee 2 weeks ago 143MB
ubuntu latest de7345b16e94 2 weeks ago 100MB
alpine latest d529dd0c6e55 6 weeks ago 8.42MB
mysql 8.0 6cd09145362d 2 months ago 799MB
sample-flask-app v1 f7d64ef4ebbb 4 months ago 141MB
myredis version1 5c08b3fad205 4 months ago 140MB
redis <none> c234a24851ba 5 months ago 140MB
hello-world latest 1b44b5a3e06a 11 months ago 10.1kB
node 18 b50082bc3670 16 months ago 1.09GB
node latest 8bbb9d1bc633 2 years ago 1.1GB
nginx 1.0 5682b6dc087e 2 years ago 169MB
ubuntu <none> ca2b0f26964c 2 years ago 77.9MB
ubuntu <none> 3db8720ecbf5 2 years ago 77.9MB
mysql <none> 019814493c7a 2 years ago 632MB
httpd latest 2776f4da9d55 2 years ago 167MB
redis <none> d1397258b209 2 years ago 138MB
docker/welcome-to-docker latest c1f619b6477e 2 years ago 18.6MB
getting-started latest 8d8a2af2207f 3 years ago 265MB
docker/getting-started latest 3e4394f6b72f 3 years ago 47MB
docker/desktop-git-helper 5a4fca126aadcd3f6cc3a011aa991de982ae7000 efe2d67c403b 5 years ago 44.2MB
C:\Users\15L>docker search ubuntu
NAME DESCRIPTION STARS OFFICIAL
ubuntu Ubuntu is a Debian-based Linux operating sys… 17862 [OK]
ubuntu/squid Squid is a caching proxy for the Web. Long-t… 129
ubuntu/nginx Nginx, a high-performance reverse proxy & we… 141
ubuntu/bind9 BIND 9 is a very flexible, full-featured DNS… 121
ubuntu/kafka Apache Kafka, a distributed event streaming … 62
ubuntu/zookeeper ZooKeeper maintains configuration informatio… 15
ubuntu/apache2 Apache, a secure & extensible open-source HT… 104
ubuntu/prometheus Prometheus is a systems and service monitori… 80
ubuntu/jre Chiseled Java runtime based on Ubuntu. Long-… 25
ubuntu/dotnet-aspnet Chiselled Ubuntu runtime image for ASP.NET a… 26
ubuntu/python A chiselled Ubuntu rock with the Python runt… 33
ubuntu/mysql MySQL open source fast, stable, multi-thread… 74
ubuntu/postgres PostgreSQL is an open source object-relation… 45
ubuntu/dotnet-runtime Chiselled Ubuntu runtime image for .NET apps… 24
ubuntu/redis Redis, an open source key-value store. Long-… 26
ubuntu/dotnet-deps Chiselled Ubuntu for self-contained .NET & A… 16
ubuntu/grafana Grafana, a feature rich metrics dashboard & … 12
ubuntu/memcached Memcached, in-memory keyvalue store for smal… 7
ubuntu/jdk Ubuntu rock for OpenJDK 0
ubuntu/rabbitmq Ubuntu-based RabbitMQ image - an open source… 0
ubuntu/loki Grafana Loki, a log aggregation system like … 2


Administrator: Command Pro
ubuntu/jdk Ubuntu rock for OpenJDK 0
ubuntu/rabbitmq Ubuntu-based RabbitMQ image - an open source… 0
ubuntu/loki Grafana Loki, a log aggregation system like … 2
ubuntu/mimir Ubuntu ROCK for Mimir, a horizontally scalab… 0
ubuntu/traefik Ubuntu ROCK for Traefik, a modern HTTP rever… 3
ubuntu/opentelemetry-collector This is a rock for the OpenTelemetry Collect… 0
ubuntu/alertmanager Ubuntu Rock for Alertmanager. It handles ale… 1
C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
What's Next?
1. Sign in to your Docker account → docker login
2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu
C:\Users\15L>docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
asd_exp3-backend latest 6c4019f6cc2c 22 hours ago 1.13GB
asd_exp3-frontend latest 22c2d416ef50 22 hours ago 62.4MB
<none> <none> cceb8c94681a 22 hours ago 1.13GB
adminer latest fed772372793 12 days ago 116MB
mysql latest 4babd5755458 13 days ago 941MB
nginx latest 4e5db4761e0f 2 weeks ago 161MB
redis latest da3958f512ee 2 weeks ago 143MB
ubuntu latest de7345b16e94 2 weeks ago 100MB
alpine latest d529dd0c6e55 6 weeks ago 8.42MB
mysql 8.0 6cd09145362d 2 months ago 799MB
sample-flask-app v1 f7d64ef4ebbb 4 months ago 141MB
myredis version1 5c08b3fad205 4 months ago 140MB
redis <none> c234a24851ba 5 months ago 140MB
hello-world latest 1b44b5a3e06a 11 months ago 10.1kB
node 18 b50082bc3670 16 months ago 1.09GB
node latest 8bbb9d1bc633 2 years ago 1.1GB
nginx 1.0 5682b6dc087e 2 years ago 169MB
ubuntu <none> ca2b0f26964c 2 years ago 77.9MB
ubuntu <none> 3db8720ecbf5 2 years ago 77.9MB
mysql <none> 019814493c7a 2 years ago 632MB
httpd latest 2776f4da9d55 2 years ago 167MB
redis <none> d1397258b209 2 years ago 138MB
docker/welcome-to-docker latest c1f619b6477e 2 years ago 18.6MB
getting-started latest 8d8a2af2207f 3 years ago 265MB
docker/getting-started latest 3e4394f6b72f 3 years ago 47MB
docker/desktop-git-helper 5a4fca126aadcd3f6cc3a011aa991de982ae7000 efe2d67c403b 5 years ago 44.2MB
C:\Users\15L>docker run -it ubuntu
root@e980db92232f:/# ^C
root@e980db92232f:/# pwd


Administrator: Command Pro
C:\Users\15L>docker run -it ubuntu
root@e980db92232f:/# ^C
root@e980db92232f:/# pwd
/
root@e980db92232f:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@e980db92232f:/# exit
exit
C:\Users\15L>docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
e980db92232f ubuntu "/bin/bash" About a minute ago Exited (0) 6 seconds ago sleepy_keller
b2b771b53cf6 asd_exp3-frontend "/docker-entrypoint.…" 22 hours ago Up 6 minutes 0.0.0.0:3000->80/tcp todo-frontend
32f8ea862963 asd_exp3-backend "docker-entrypoint.s…" 22 hours ago Up 6 minutes 0.0.0.0:5000->5000/tcp todo-backend
e70089016f33 mysql:8.0 "docker-entrypoint.s…" 2 days ago Created mysqlserver
df9f3956f856 nginx "/docker-entrypoint.…" 2 days ago Exited (255) 23 hours ago 80/tcp, 0.0.0.0:3307->3306/tcp webserver
290c69b37455 redis "docker-entrypoint.s…" 7 days ago Exited (0) 7 days ago t2_database
2c7139eeda3f ubuntu "/bin/bash" 7 days ago Exited (0) 7 days ago boring_zhukovsky
8ddb1e79e05f ubuntu "/bin/bash" 9 days ago Exited (0) 9 days ago distracted_ishizaka
298abd2b1a9d c234a24851ba "docker-entrypoint.s…" 4 months ago Exited (0) 4 months ago naughty_shirley
1a197c2bd0f5 c234a24851ba "docker-entrypoint.s…" 4 months ago Exited (255) 9 days ago 6379/tcp happy_galois
013ed608bab8 c234a24851ba "docker-entrypoint.s…" 4 months ago Exited (0) 4 months ago nifty_yonath
f9b579837c27 d1397258b209 "docker-entrypoint.s…" 4 months ago Exited (0) 4 months ago vigilant_knuth
06b8f6ece314 ca2b0f26964c "/bin/bash" 2 years ago Exited (255) 4 months ago happy_cerf
7b8921bf4fb0 d1397258b209 "docker-entrypoint.s…" 2 years ago Exited (255) 4 months ago 6379/tcp jatinn
c782c1137761 node "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago jatingupta
ca1d141bfabd node "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago reverent_williams
7638b03139a0 d1397258b209 "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago jatin
d53dc9583439 d1397258b209 "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago happy_engelbart
45c72eb69cd7 d1397258b209 "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago keen_blackburn
b9b0d800f902 d1397258b209 "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago priceless_jackson
a91dfde56008 d1397258b209 "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago wizardly_sanderson
87fe33760977 d1397258b209 "docker-entrypoint.s…" 2 years ago Exited (0) 2 years ago epic_burnell
8823c8213441 ca2b0f26964c "/bin/bash" 2 years ago Exited (0) 2 years ago AHMED
95a79a09eca1 ca2b0f26964c "/bin/bash" 2 years ago Exited (0) 2 years ago festive_chatelet
aefcc9a16a00 ca2b0f26964c "/bin/bash" 2 years ago Exited (0) 2 years ago festive_kare
2b923e117ec0 docker/welcome-to-docker:latest "/docker-entrypoint.…" 2 years ago Exited (0) 4 months ago welcome-to-docker
29dadd522f4b docker/getting-started:latest "/docker-entrypoint.…" 2 years ago Exited (0) 2 years ago kind_bhaskara
cfb503a59d94 nginx:1.0 "/usr/sbin/nginx -g …" 2 years ago Exited (0) 2 years ago xyz
d2bc0c136fb2 httpd "httpd-foreground" 2 years ago Exited (255) 2 years ago 80/tcp busy_edison
2e821f5cfabb 3db8720ecbf5 "/bin/bash" 2 years ago Exited (0) 2 years ago boring_diffie
0de78c8adee9 getting-started "docker-entrypoint.s…" 3 years ago Exited (255) 3 years ago 0.0.0.0:3000->3000/tcp frosty_tharp
38f4628b6bf4 docker/getting-started "/docker-entrypoint.…" 3 years ago Exited (255) 3 years ago 0.0.0.0:80->80/tcp youthful_neumann

Conclusion: In conclusion, this experiment successfully demonstrated the practical application of containerization and environment isolation using the Docker framework. By managing official image repositories, executing interactive and detached container lifecycles, and establishing host-port mapping, the core competencies of consistent application deployment outlined in Learning Outcome 4 (LO4) were achieved. Furthermore, simulating container resource monitoring and system inspections underscored the importance of environment reproducibility and infrastructure immutability within collaborative Agile and DevOps engineering environments. LO4 mapped successfully.